



Arduino Based GSM HOME SECURITY SYSTEM

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This project is designed using normally-closed reed switches connected to doors and windows and additional passive infrared (PIR) motion sensors to detect movement of a burglar or an unwanted intruder in your home. The security system can dial up to two phone numbers to alert you. It can also send text message to one of the numbers.

Circuit and working

Fig. 1 shows circuit diagram of the Arduino based GSM home security system. It works off a 12V power supply (connected across CON2). An Arduino microcontroller (MCU) requires only 5V but a GSM modem requires 12V. As Arduino Uno has an inbuilt 5V voltage regulator, a common 12V supply can be used for the whole system. It is built around two Arduino Uno boards (Board 1) and (Board 2), one SIM900 GSM module, TIP142 (T1, T2) and 2N2222 (T3-T5) transistors, magnetic reed switches, PIR motion sensors and a few other components.

Brain of the circuit is Arduino Uno MCU board (Board 1). PIR sensors (connected across CON4 through CON8) detect motion, and output of the PIR sensor goes high when it detects any motion. The software constantly scans for a positive input at pin 9 (D9) of Board 1. When all the reed switches are closed and outputs of all PIR sensors are low, D9 is held low by T5.

A positive output on pin 8 (D8) or pin 11 (D11) of Board 1 is necessary to put the system in armed

mode, which is indicated by LED1. As soon as one of the reed switches in the closed loop opens, or PIR sensors detect motion, logic level on D9 goes high via D8 or D11.

Pin 10 (D10) of Board 1 also goes high and latches D9 to 5V. Board 1 then commands SIM900 GSM module to send a text message and make six phone calls each to two different mobile numbers.

PARTS LIST

Semiconductors:	
Board 1, Board 2	- Arduino Uno
T1, T2	- TIP142 npn transistor
T3-T5	- 2N2222 npn transistor
D1-D4, D6-D10	- 1N4148 signal diode
D5	- 1N4001 general-purpose plastic rectifier
LED1-LED3	- 5mm LED
Resistors (all 1/4-watt, $\pm 5\%$ carbon, unless stated otherwise):	
R1	- 2.7-kilo-ohm
R2	- 680-ohm
R3	- 47-kilo-ohm
R4, R7	- 1-kilo-ohm
R5, R9, R10	- 2.2-kilo-ohm
R6	- 100-kilo-ohm
R8, R15	- 10-kilo-ohm
R11, R12	- 68-kilo-ohm
R13	- 220-ohm
R14	- 470-ohm
Capacitors:	
C1, C7	- 2200pF, 40V electrolytic
C2-C6, C8	- 100nF ceramic disk
Miscellaneous:	
S1, S2	- Push-to-on switch
S3-S6	- Magnetic reed switches
L1, L2	- 100 μ H moulded choke
RL1	- 12V, 1C/O relay
CON1	- 6-pin connector terminal for SIM900 GSM module
CON2	- 2-pin connector terminal for 12V DC
CON3	- 2-pin connector for external siren
CON4-CON8	- 3-pin connector for PIR sensors
CON9	- 4-pin connector for Board 2
CON10	- 14-pin connector for Board 1
	- SIM900 GSM module
	- Siren
	- 12V battery or 12V, 2A DC regulated power supply
	- HCSR501 motion sensor X5

You can simply ignore the calls and call your neighbour or a security guard to check on your home or premises.

After making a total of 12 phone calls, pin 13 (D13) of Board 1 goes high. It instructs Board 2 to cut off power supply to Board 1 and GSM module for five seconds. T1 switches off and on, thus resetting the system. Board 1 returns to standby mode, which is indicated by LED2.

The system can be activated again by sending an SMS. Each time the circuit is powered up, LED2 lights up to indicate that the system has entered standby mode. It also sends an SMS to your mobile phone that reads 'Standby Mode.'

You can arm the system by sending SMS '\$ON\$' and the circuit would reply back 'Alarm On.' The system now monitors all reed switches in the closed loop and output from all PIR sensors. It can be disarmed by sending SMS '\$OFF1234\$'. It would reply back 'Alarm Off.'

A four-digit password is needed to disarm the alarm system by SMS. It can easily be changed by modifying the line in source code of Arduino1.ino sketch as shown below for reference.

```
if (inchar == '0' || inchar == 'F' || inchar == 'f' || inchar == '1' || inchar == '2' || inchar == '3' || inchar == '4')
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Before leaving home, you can also press push button S1 for 30 seconds instead of arming the alarm by sending an SMS. LED2 would start flashing. After 30 seconds delay, LED2 stops flashing